

Effect of Adjuvant Yoga Therapy for Chronic Respiratory Diseases in Pediatric Population – A Scoping Review

Abstract

Chronic respiratory diseases (CRD), i.e., asthma, COPD, allergic rhinitis, and tuberculosis, account for approximately one-third of emergency hospitalizations. As a holistic therapy, yoga has the potential to improve both the physical and psychological well-being of individuals with CRD, and its popularity has expanded globally. However, studies assessing yoga's effects have yielded inconsistent results. The scoping review adhered to the PRISMA Extension for scoping reviews, PRISMA-ScR. A protocol was registered on OSF (10.17605/OSF.IO/Q6CKN). We conducted a systematic search on PubMed, Embase, and Scopus, covering studies from inception till december-2024. Inclusion criteria were a pediatric population with CRD, yoga as adjuvant therapy, and English language publications. Data were extracted systematically and categorized based on publication year, author, type of study, country, disease, and intervention. We included 552 database results with 2357 patients, published between 1985 and 2023. Most studies were primarily from India ($n = 15$), focused on asthma, along with two studies on cystic fibrosis and one each on pleural effusion and bronchiectasis. Study types included observational studies ($n = 10$), randomized controlled trials ($n = 8$), controlled trials ($n = 1$), quasi-experimental studies ($n = 1$), and pilot studies ($n = 2$). Yoga improved pulmonary function test scores in asthma, pleural effusion, and bronchiectasis, enhancing quality of life (QOL) and psychological well-being. However, no studies were found on COPD, pneumonia, and tuberculosis in the pediatric population. Quality evidence suggests that yoga improves asthma symptoms and QOL. However, variations in yoga modules limit generalizability. Future research should develop validated disease-specific modules, including unexplored facets like Yama, Niyama, yogic counseling, and diet. Mechanistic studies with long follow-up periods are needed.

Keywords: Asthma, cystic fibrosis, pulmonary function test, Yoga

Introduction

Chronic respiratory disease (CRD) is an umbrella term that refers to a broad range of chronic respiratory conditions. It impacts over 1 billion people worldwide with infants and young children being especially susceptible to it.^[1] The five major respiratory conditions that contribute to global health burden are asthma, allergic rhinitis, cystic fibrosis, chronic obstructive pulmonary disease, and tuberculosis. These are not only the third leading cause of mortality globally in 2019, but also associated with a lot of financial burden and cost. There is a variable and constant increase in the prevalence of asthma worldwide in the last two decades.^[2] However, the greater strides have been made over the past two decades in reducing under five mortality rate from 90 per 1000 live births in 1990 to 48 per 1000 live births in 2012.^[3]

Allergic respiratory disorders such as asthma and allergic rhinitis are more prevalent among children attending pediatric clinics, with a prevalent rate of about 14% and 23.5%, respectively. A recent report (2022) suggests that the prevalence of allergic rhinitis in 6–7-year-old children was 7.7% and 23.5% in children aged 13–14 years. A recent report (2022) suggests that the prevalence of allergic rhinitis in 6–7-year-old children was 7.7% and 23.5% in children aged 13–14 years. Unlike other diseases, the prevalence of these disorders is higher in developed countries as compared with lower and middle-income countries; however, because of poor facilities and quality of care, the percentage of severe asthma is as high as 30% with 80% of asthma-related deaths in lower and middle economic countries.^[4,5] Cystic fibrosis is a common

**Monica Mahendru¹,
Sriloy Mohanty²,
Dheeraj Dabas³,
Vivek Singh
Malik^{4,5,6},
Meenu Singh^{4,5,6}**

*Departments of ¹Pediatrics,
²Ayush, ³Telemedicine and
Biomedical Informatics,
⁴Evidence Synthesis and
⁵Telemedicine and Biomedical
Informatics, AIIMS, ⁶Cochrane
Evidence Synthesis Unit, AIIMS,
Rishikesh, Uttarakhand, India*

Address for correspondence:

Prof. Meenu Singh,
All India Institute of
Medical Sciences, Rishikesh,
Uttarakhand, India.
E-mail: espcentre@aiimsrshikesh.edu.in

Access this article online

Website: <https://journals.lww.com/IJOY>

DOI: 10.4103/ijoy.ijoy_147_25

Quick Response Code:



This is an open access article distributed under the terms of the Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 License (CC BY-NC-ND), where it is permissible to download and share the work provided it is properly cited. The work cannot be changed in any way or used commercially without permission from the journal.

For reprints contact: WKHLRPMedknow_reprints@wolterskluwer.com

How to cite this article: Mahendru M, Mohanty S, Dabas D, Malik VS, Singh M. Effect of adjuvant yoga therapy for chronic respiratory diseases in pediatric population – A scoping review. *Int J Yoga* 2026;19:128-38.

Submitted: 04-Jun-2025

Revised: 05-Nov-2025

Accepted: 12-Nov-2025

Published: 25-Jun-2026

inherited life-limiting disease, with its estimated prevalence in India to be around 1 in 43,321–1 in 100,323.^[6] Similarly, Tuberculosis is also a major public health concern globally, and pediatric tuberculosis contributes to 27% of total cases and 8%–20% of Tb-related deaths in India.^[7]

Despite the availability of various evidence-based conventional pharmacological and nonpharmacological interventions for chronic childhood respiratory disorders, it is observed that the prevalence slope is steadily elevating, especially in lower and middle economic countries. Thus, pulmonary rehabilitation (PR) is one of the major front-line add-on programs recommended across different societies, including the British Thoracic Society.^[8] The structure of PR includes key components such as exercise training, health education, behavior modification, nutritional counseling, and breathing technique. In high-income countries, these components have been shown to improve quality of life (QOL), reduce morbidity, and improve pulmonary function. However, there are a number of important barriers that prevent PR from being implemented effectively in low-socioeconomic environments. These include scarce resources, including a lack of PR experts and tools, neither culturally acceptable nor covered in insurance, as well, a lack of knowledge about PR initiatives and practical difficulties such as the expense of transportation, conflict with patients' work schedules. These barriers highlight the urgent need for the development of evidence-based, culturally acceptable, and cost-effective PR programs that can be feasibly implemented in resource-constrained environments.^[9] In the past two decades, different nonpharmacological rehabilitation tools such as exercise, Tai-chi, Pilates, and yoga have been used. Therefore, Yoga, being an ancient Indian, behavioral, mind–body intervention for holistic health dimension, its subdomains are yoga postures (asana), breathing exercises (Pranayama), cleansing practices (shatkarma), dietary habits (ahara), and behavioral modifications (vihara), ethical, social, and personal conduct (Yama and Niyama) along with counselling. Integrating these practices, yoga has emerged as evidence-based mind–body medicine for enhancing outcomes and QOL among patients with pulmonary disorders.^[10] Yoga studies have shown significant improvement in pulmonary functions, inspiratory muscle strength, and overall QOL in various respiratory diseases.^[11,12] However, the scarcity of evidence is available in regards to the efficacy of Yoga as an adjuvant therapy among the pediatric population.

This review aims to conduct a scoping review of all the studies where yoga was given as an adjuvant intervention among the pediatric population with CRDs and attempt to develop a yoga-based PR model for the aforementioned population at the end of the review.

Methods

Protocol and registration

The scoping review was conducted in accordance with PRISMA Extension for scoping reviews (PRISMA Scr).^[13] A

protocol was prepared prior to the data extraction process and was registered in OSF (<https://doi.org/10.17605/OSF.IO/Q6CKN>). The review was conducted in various steps, such as developing a research question, selecting studies based on inclusion and exclusion criteria, charting the data, and summarizing and reporting the results.

Data sources and search strategy

Two researchers, MM and DD, conducted the systematic search across four electronic databases, including PubMed, Embase, Scopus, and Google Scholar from the inception till March 2024, using various MeSH terms such as “Yoga, bronchial asthma, Cystic fibrosis, Pneumonia, COPD, Allergic rhinitis” [Supplementary Table 1] in consultation with MS and SM. The search terms were deliberately kept broad, and no limits were applied to publication date as well. A forward and backward search was also conducted to include relevant studies. The focus was on studies where yoga was provided as an adjuvant therapy for individuals with CRDs. This review included all types of interventional studies and excluded review commentaries and cross-sectional studies.

Papers were selected for full text screening after screening based on title and abstract. Studies were screened based on various criteria: A – Pediatric Population, B – Having chronic respiratory disorder such as asthma, chronic obstructive pulmonary disorder, allergic rhinitis, pneumonia, allergic rhinitis, C – Studies in which yoga was given as an adjuvant therapy in the treatment of these disorders. D – Published in English Language.

Study selection

Two reviewers (MM, DD) were responsible for searching, screening, and reviewing articles. Discrepancies were resolved by the third reviewer (SM). Microsoft Excel was used for recording the data and selecting research studies. The articles containing Yoga as an adjuvant intervention along with chronic respiratory disorders in the pediatric population were manually identified. After initial screening based on title, inclusion and exclusion criteria were applied, and studies with available full text were selected.

Data extraction and synthesis

Data were extracted in a standardized manner, including segregation based on year of publication, author, type of study and country, disease, and intervention. Out of 16 studies from PubMed, 8 studies were retrieved from PubMed after removing duplicate studies, irrelevant studies based on title and abstract, and full text screening. Out of 14 studies from Embase, 4 studies were retrieved as final studies. Out of 10 studies from Google Scholar, 8 were retrieved as final for inclusion in the scoping review. The full text of all articles with Yoga as an adjuvant intervention was picked manually. The final studies included were published from 1985 to 2024, including randomized controlled trials, pilot trials, and quasi-experimental trials.

The details of screening and inclusion, exclusion criteria are depicted in the PRISMA Flow chart.

Results

Following the Initial search, 552 results were retrieved. After removing 454 duplicates and 61 studies based on title and abstract, 37 studies were sought for retrieval and screened. During the screening process, attention was paid to articles having yoga as an intervention, and was carried out on the pediatric population with pulmonary diseases. Out of these 37 studies, 15 studies were excluded based on the age of participants, articles in other languages, or full-text not available. Finally, 22 studies were included in the scoping review. The studies had a common intervention as yoga, and outcomes were different based on different pulmonary diseases, such as asthma, cystic fibrosis, COPD, tuberculosis, and allergic rhinitis. The PRISMA Flow diagram for the study selection process is shown in Figure 1.

Yoga for asthma

Bronchial asthma is a chronic pulmonary disease which results in spasm of the smooth muscles of the airway, leading to obstruction of bronchioles, hence causing recurrent episodes of shortness of breath, coughing, wheezing, and chest tightness.^[14] Yoga exercises along with pranayama, increase chest wall expansion, lung compliance, enhance the strength of respiratory muscles, diaphragm, and upper abdominal muscles, and improve respiratory breathing capacity.^[15]

Yoga has been employed in numerous trials and has shown a positive impact on asthma management among adolescents and children. In this review of 13 studies involving 1085 patients who underwent yoga interventions, including asanas, pranayama, meditation, consistently demonstrated improvements in pulmonary function tests (PFTs) parameters such as Forced Vital

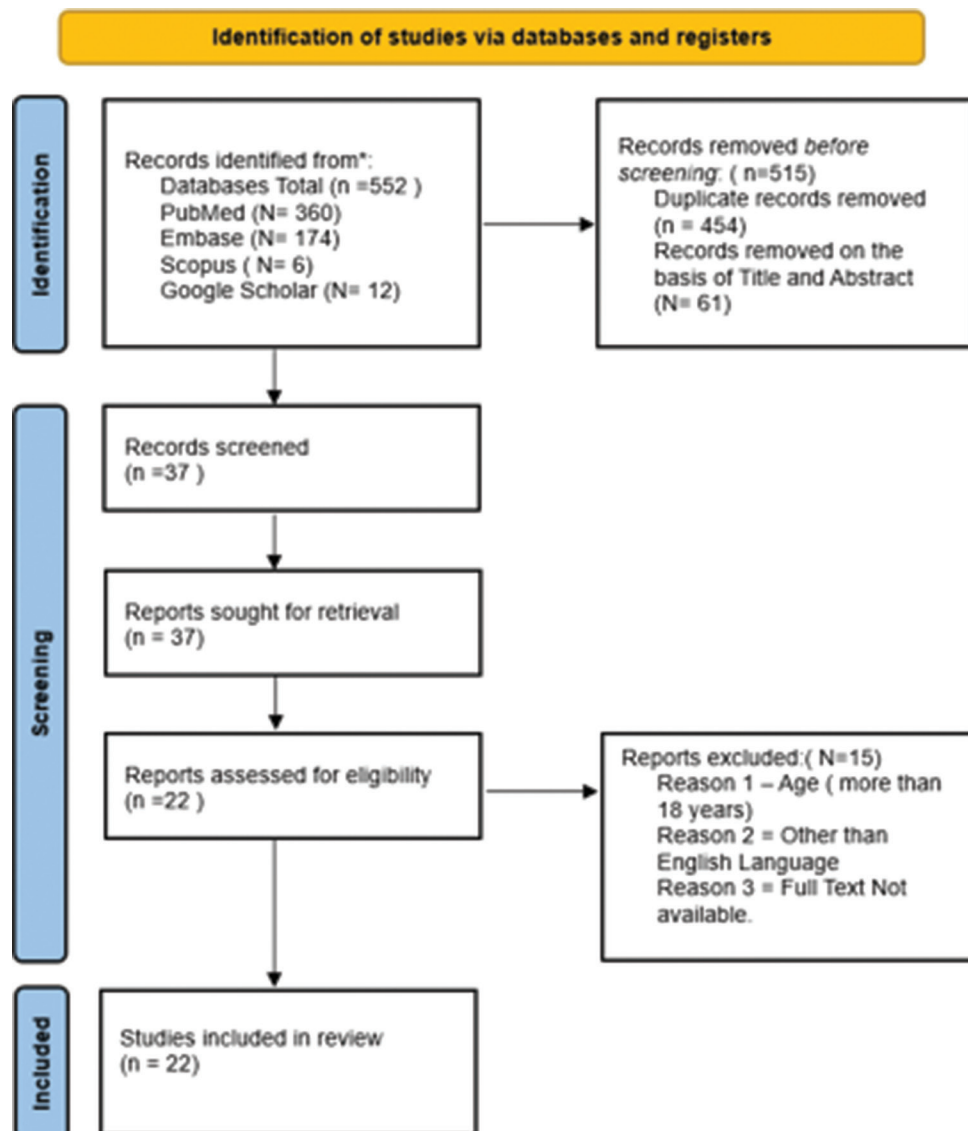


Figure 1: PRISMA flow chart of the studies

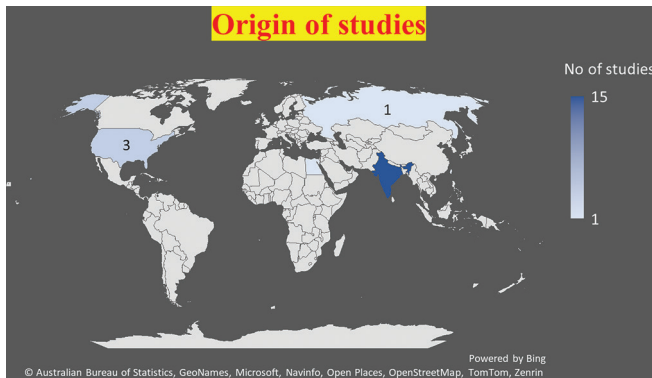


Figure 2: This figure represents 22 studies, out of which fifteen studies were from India ($n = 15$), one from Russia ($n = 1$), one from Ethiopia ($n = 1$), one from Taiwan ($n = 1$), one from Egypt ($n = 1$), and five from the USA ($n = 3$)

Capacity (FVC), forced expiratory volume in 1 s (FEV_1), $FEV_1/FVC\%$, peak expiratory flow rate, as compared to the control arm. For instance, studies by Yadav *et al.*^[16] and Yadav *et al.*^[17] reported significant improvements in lung function and reduction in asthma symptoms [Figure 2].

QOL was used as a parameter in 6 studies involving 563 patients, who had their QOL assessed, representing 25.59% of the total population.^[18] Psychological benefits were also observed in 2 studies ($n = 2$) which included reduction in anxiety, improvement in various scores such as symptom score, activity limitation score, emotional functional score, and total PAQLQ score, as observed in studies by Jasrotia *et al.*^[19] In the study by Yadav P *et al.*,^[16] yoga interventions led to notable improvements in QOL scores. Preintervention – the activity limitation score was 5.82 ± 0.61 which improved to 5.63 ± 0.69 . The emotional functioning score got improved from 5.67 ± 0.40 to 5.39 ± 0.52 . The intervention group showed a notable reduction in illness severity score, fewer weekly asthma attacks, and fewer school days lost per year as compared to the control group. Moreover, improvement in QOL was observed in $n = 3$ studies conducted by Nasrollahpor *et al.*^[20] Pandya.^[21] In the study by Nasrollahpor *et al.*, it was found that breathing exercises such as yogic complete breathing, nadishodhan, and pranayama, if performed 10–15 min daily for 8 weeks, improved the QOL in patients in the intervention group as compared to the control group, likely by modifying their breathing patterns. They observed that frequent exercise might be necessary to further enhance the QOL, as evident from their study, where children practiced these exercises daily for 2 months. The PedsQL 3.0 asthma module was utilized to measure these improvements. Another study by Pandya *et al.* demonstrated that following the implementation of the Yoga Education Program, children in the intervention group exhibited better asthma control. The study represented lower average pediatric asthma diary scores and higher childhood asthma control test (C-ACT) scores as compared to the control group.

However, a study by Sodhi *et al.* was not considered due to a mixed group population (17–50 years) which involved

120 patients with asthma and were randomly divided into two groups, i.e., Group A (Yoga), Group B (Control group). Both the groups received medications but the difference was that the yoga group practiced yoga breathing exercises for 45 min per day for 8 weeks. The results showed that Yoga Breathing Exercise improved the environmental, activity, and symptoms subscale of the Asthma Quality of Life Questionnaire.^[22,23]

In conclusion, our study along with other referenced research, suggests that yoga can improve QOL and PFTs through breathing regulation mechanisms. All studies showing the effect of yoga in patients with bronchial asthma are summarized in Table 1.

Yoga for cystic fibrosis

While Yoga has demonstrated beneficial effects in patients of bronchial asthma, its impact on patients with cystic fibrosis remains relatively underexplored. The study by Ruddy *et al.*^[24] involved 30 adolescents and young adults aged 12–25 years, who were randomized into two groups: an intervention group participating in a 12-week yoga program and a control group. The intervention group exhibited significant improvements in QOL, as assessed by the Cystic Fibrosis Questionnaire Revised. The 12-week Yoga program was deemed safe and well-tolerated among participants with mild to moderate lung disease. Another study by McNamara *et al.*^[25] investigated the effect of yoga therapy on anxiety and joint pain in children with cystic fibrosis. Participants aged 8–16 years underwent a 6-week yoga program with sessions held twice a week. Results revealed a reduction in anxiety scores from 40.5 to 28.3, as measured by the State-Trait Anxiety Inventory for children, and a decrease in joint pain scores from 4.2 to 2.1, as measured by the Wong–Baker FACES pain rating scale.

The studies are summarized in Table 2.

Yoga for bronchitis

Chronic inflammation in the bronchi, accompanied by obstructive symptoms in adolescents, compromises the sensitivity of the muscles of the bronchial tree, leading to increased hyperresponsiveness to external stimuli. Given the high incidence of chronic bronchitis in this population, it is crucial to explore the potential of yoga exercise as a rehabilitative intervention for adolescents with chronic obstructive bronchitis. For instance, a study by Medvedev *et al.*^[26] demonstrated the improvement in lung function, i.e., FEV_1 increased from 65% to 75% of the predicted value, a reduction in respiratory rate from 24 to 20 beats/min, improvement in oxygen saturation levels from 92% to 96% with improved exercise tolerance. These findings underline the potential effects of yoga, but there is a need for more studies exploring improvement in pulmonary functions, clinical outcomes, and QOL. The studies are summarized in Table 2.

Table 1: It summarizes the studies showing the effect of yoga in children diagnosed with bronchial asthma

Author/ year	Country	Study design	Intervention	Study duration	n	Age (years)	Disease condition	Parameter	Test	Control
Nagarathna <i>et al.</i> (1985)	India	RCT	Breathing exercises (5 min) - Yogasanas-(a) General yogasanas (20 min). Simple physical postures (in the standing, sitting, prone, and supine positions) Pranayama (10 min, meditation and devotional session) (15 min, neti and vaman dhouti) followed by Savasana	2 weeks	106	9–47	Asthmatics	Number of attacks of asthma/ week Asthma Severity score	0.83±2.49 0.75±0.8	2.1±2.7 1.05±0.85
Jain SC <i>et al.</i> (1991)	India	Single- group pre– post study design	(i) Yogic cleansing procedures (Kriyas); (ii) maintenance of yogic body postures (Asanas) (6, 7), and (iii) yogic breathing practices (Pranayamah)	6 weeks	46	11–18	Asthmatics	FEV ₁ (% pred) MVV (L/min)	Pre 67.9±15.1 70.9±19.7	Post 75.8±13.9 85.8±21.9
Khanam <i>et al.</i> (1996)	India	Single group design	Yoga Asanas (Padmasana, Tadasana, Utkatasana, Chakrasana, Trikonasana, and Bhujangasana), Pranayamas (voluntary regulation of respiration), Nadi Shodhan, Bhramari, Anuloma Vilomas, Preksha Dhayan, and lectures on the philosophy of yoga - 14 sessions for 7 days	1 weeks	9	12–60	Asthmatics	FVC FEV ₁ PEFR	Pre 2.10±0.65 1.51±0.34 5.79±1.20	Post 1.73±0.41 1.45±0.39 6.5±1.65
Sodhi <i>et al.</i> (2008)	India	RCT	Pranayama, kapalabhati, bhastrika, ujjayi	8 weeks	E=60 C=60 T=120	17–50	Asthmatic	FEV ₁ FVC PEFR	83.16±10.49 86.6±10.72 82.45±10.75	77.26±9.86 83.3±10.00 79.42±8.26
Chiang <i>et al.</i> (2009)	China	RCT	Relaxation-breathing training	12 weeks	E=22 C=26 T=48	6–14	Asthmatics	Asthma signs/ symptoms PEFR Children's anxiety	6.93±1.51 282.50±76.72 26.11±11.41	7.24±1.71 240.12±63.62 1.94±2.18
Mekonnen <i>et al.</i> (2010)	Eithopia	Non-RCT	Integrated yogic practice, Yogic Postures, Breath slowing technique (10 min), Discussion (10 min) to understand the emotional well- being and positive outlook	4 weeks	E=12 C=12 T=24	12–50	Asthmatics	Change in drug use - Salbutamol tablet Salbutamol puff	Reduced - 7 (58.3) No change - 2 (16.7) Increased - 3 (25) Reduced - 8 (66.7)	Reduced - 1 (8.3%) No change - 3 (25%) Increased - 9 (66.7%) Reduced - 2 (16.65%)

Contd...

Table 1: Contd...

Author/ year	Country	Study design	Intervention	Study duration	n	Age (years)	Disease condition	Parameter	Test	Control
									No change - 3 (25)	No change - 8 (66.7)
									Increased - 1 (8.3)	Increased - 2 (16.65)
Jasrotia <i>et al.</i> (2013)	India	RCT	Sukshma Vyayama (Micro exercises) (Macro exercises) Asanas (Postures), Pranayama, Dharna, and Dhyana (Om meditation)	12 weeks	E=15 C=15 T=30	10–14	Asthmatic	VC FVC FEV ₁ FEV ₁ (%) PEFR MEFR	2.26±0.82 2.34±0.72* 1.73±0.65 81.46±9.97 4.09±1.25 1.93±0.72	1.97±0.50 1.88±0.54 1.53±0.54 81.05±13.42 3.37±0.92 1.91±0.93
Agnihotri <i>et al.</i> (2016)	India	RCT	Asanas (Gomukhansan, Ardhamatsyendrasana, Paschimottanasana, Bhujangasana, Dhanurasana, Na, Parvatasana, Tadasana, Shavasana), Pranayama (Nadisho dhana, Bhastrika, Bhramar), Meditation	6 months	E=138, C=138, T=276	12–60	Asthmatics	FVC FEV ₁ FEV ₁ /FVC PEFR	85.83±8.07 78.39±4.55 66.62±3.97 75.62±4.71	81.98±5.32 65.18±3.64 70.36±4.92 65.08±5.21
Gupta <i>et al.</i> (2017)	India	RCT	Nadishodhana and Ujjayi Pranayama, and yoga postures like Virasana, Suptasana, and Supta Baddhakonassna Tadassna, Bhujangassna Gomukhassna, Shavassna, and Makarassna	6 weeks	E=75 C=75 T=150	8–14	Asthmatics	FVC FEV ₁ PEFR MVV FEV ₁ /FVC (%)	77.30±17.09 74.30±15.36 76.68±15.81 57.92±14.20 96.48±4.42	75.32±18.60 71.42±17.81 71.60±15.63 52.32±11.57 94.85±4.88
Ramadan Azab <i>et al.</i> (2017)	Egypt	RCT	Sukha Purvaka Pranayama, Dirgha Pranayama, Yoga chair breathing	12 weeks	E=20 C=20 T=40	7–12	Asthmatics	FEV ₁ FVC FEV ₁ /FVC (%) RV 6MWT	89.2±10.5 96.7±8.2 86.2±10.8 87.3±16.2 435±65.2	90.2±12.1 98.8±8.6 86.9±10.3 88.3±17.2 440±66.2
Agnihotri Shruti <i>et al.</i> (2017)	India	RCT	Asanas (Gomukhansan, Ardhamatsyendrasa, Paschimottanasana, Bhujangasana, Dhanurasana, Na, Parvatasana, Tadasana, Shavasana) Pranayama (Nadishodhana, Bhastrika, Bhramar), meditation	6 months	E=125 C=130 T=255	12–60	Asthmatics	QOL	4.92±0.38	4.79±0.56
Jasrotia <i>et al.</i> (2019)	India	RCT	Sukshma Vyayama, Sthula vyayama, Asana, Pranayama, Dharna, Dhyana	12 weeks	E=15 C=15 T=30	10–14	Asthmatics	Illness score Severity score	4.93±2.09 1.38±0.29	6.60±1.88 1.52±0.21

Contd...

Table 1: Contd...

Author/ year	Country	Study design	Intervention	Study duration	n	Age (years)	Disease condition	Parameter	Test	Control
								Asthm attack/week	1.78±0.39	2.01±0.89
								School day lost/year	21.83±11.84	27.45±16.32
Pandya samta (2019)	India		Yoga education program	1 year	E=450 C=450 T=900	11–18	Asthmatics	C-ACT PAD	24.04±3.65 4.82±2.76	12.03±5.11 14.17±2.42
Lack Sharon <i>et al.</i> (2020)	USA	Pilot single group	Mindful yoga intervention	1 session	T=15	7–11	Asthmatic	FEV ₁ FVC FEV ₁ /FVC (%)	83.73±20.21 91.33±16.31 80.87±8.76	82.07±24.31 87.73±19.56 81.00±10.91
Yadav Pallavi <i>et al.</i> (2021)	India	RCT	Tadasana, Katichakrasana, Gomukhasana, Bhujangasana, Savasana, and pranayama Nadishodhan and Kapalbhati, and meditation	12 weeks	E=70 C=70 T=140	10–16	Asthmatics	FVC FEV ₁ FEV ₁ /FVC PEFR PAQLQ	99.29±8.87 101.07±10.69 101.26±7.29 94.50±15.92 5.67±0.40	95.61±9.14 94.36±9.91 97.44±6.91 90.99±16.83 5.39±0.52
R mahboobeh <i>et al.</i> (2021)	India	Quasi experimental	Padmasana, Yoga Mudra, Matsyasana, Kukkutasana, Uthana Padasana, Pavanmuktasana, Paschimottanasana; Dhanurasana, Supta Vajrasana, Gomukhasana, Viparita Karani, Sarvangasana, Halasan, Karnapidasana, Bhujangasana, Bakasana, Mandukasana, Parvatasana, Nauli and Shavasana	8 weeks	E=15 C=15 T=30	8–14	Asthmatic	FEV ₁ FEV ₁ /FVC	94.80±2.59 96.40±2.94	90.13±5.01 94.80±4.03
Yadav Rashmi <i>et al.</i> (2023)	India	RCT	Yoga + standard treatment Bhramari pranayama, om chanting	12 weeks	E=56 C=54 T=110	8–15	Asthmatic	FVC FEV ₁ FEV ₁ /FVC PEFR FEF 25:75	0.23±0.61 1.84±0.52 78.82±12.70 4.32±1.44 3.46±1.38	0.24±0.62 1.74±0.54 80.43±11.85 4.14±1.19 3.40±1.21
N Somayeh <i>et al.</i> (2024)	Iran	Quasi RCT	Complete yogic breathing (abdominal, intercostal, clavicular), kapalbhati breathing, Nadishodhna breathing	8 weeks	E=42 C=42 T=84	5–11	Asthmatic	QoL	79.4±9.1	72.7±11.1

FEV₁: Forced expiratory volume in one second, FVC: Forced vital capacity, PEFR: Postexpiratory flow rate, RV: Residual volume, 6MWT: 6 min walk test, MVV: Maximum voluntary ventilation, MEFR: Mid expiratory flow rate, PAQLQ: Pediatric Asthma Quality of Life Questionnaire, C-ACT: Childhood asthma control test, PAD: Pediatric asthma diary, RCT: Randomized controlled trial, FEF: Forced expiratory flow, QoL: Quality of life

Yoga for pleural effusion

Yoga has been shown to alleviate symptoms associated with pleural effusion, improve lung function, and enhance QOL, thereby reducing anxiety and stress. The study demonstrated a 30% reduction in symptom severity in the interventional group as measured by the standardized symptom severity scale. Moreover, the intervention group exhibited a 12% increase in FEV₁ compared to the control arm. QOL index scores also improved significantly with an average increase from 55 to 72.^[27] These findings underscore the potential benefits of yoga in this context, highlighting the need for further studies with a large sample size and more robust methodology to confirm these outcomes. The studies are summarized in Table 2.

Discussion

This scoping review aimed to determine the effect of Yoga as an intervention in the pediatric population with chronic respiratory disorders. In this current review, 20 studies were found after conducting a comprehensive search across various databases. While there are a significant number of

studies exploring the effect of yoga on asthma but very little evidence is obtained for cystic fibrosis ($n = 2$), pleural effusion ($n = 1$), bronchiectasis ($n = 1$), and no studies for COPD and pneumonia were found out on the pediatric population. Therefore, this review opens up new ideas for research and proper modules that need to be developed on the same. However, expanding the research scope to include these conditions is essential for understanding the potential benefit of yoga in pediatric respiratory care [Figure 3].

Limitation

Although the yoga modules in various studies incorporated similar facets of yoga practices such as asanas, loosening, and pranayama, surprisingly, the yoga practices in these studies vary from each other. We would like to note that with different yoga modules, the generalization of the effect of yoga may be difficult. Future research should focus on developing and validating yoga modules of different diseases. Even though there were studies of shorter duration or follow-ups which limit the long-term therapeutic benefits of yoga, and even though there is scarce of evidence exploring the possible mechanism behind the beneficial

Table 2: Summarizes the studies showing the effect of yoga in patients with cystic fibrosis, bronchitis, and pleural effusion

Author (year)	Country	Study design	Intervention	Study duration	n	Age (years)	Disease condition	Parameter	Test	Control
Prakasamma et al. (1983)	India	Non-RCT	Pranayama	20 days after fluid aspiration	10	16–50	Pleural effusion	FVC FEV ₁ PEFR	483 233 113	348 249 86
Ruddy et al. (2015)	USA	Single-group pilot study	Breathing exercises (3–5 min), Postures (35–40 min), Relaxation and final breathing (10–15 min)	8 weeks	10	12–25	Cystic fibrosis	Post-CFQ-R respiratory domain CFQ-R physical domain	67.9±11.4 88.4±9.5	82.1±9.9 88.9±18.8
Namara Mc et al. (2016)	SA	Single group design	Selected yoga posture	10 weeks	12	7–21	Cystic fibrosis	Pre, median (range) STAIC Respiratory symptoms Pain, n (%) Sleep quality HADS anxiety HADS, n (%) Depression Abdominal pain Joint pain	Post, median (range) 28 (19–39) 79.2 (38.9–100) 7 (35) 8.5 (0.15) 6.5 (0.12) 4 (1.4) 4 (1.4)	26 (20–31) 66.7 (38.9–100) 7 (35) 9 (1.15) 8 (0.12) 3.5 (1.4) 4 (2.4)
Medvedev et al. (2021)	Russia	Non-RCT	Cobra, bow, locust, folding knife, birch, cleansing, breath, forging bellows, noisy breathing, rhythmic breathing, vacuum breathing	16 weeks	E=24 C=22 T=46	14–15	Bronchitis	FVC FEV ₁	2.3±0.17 2.2±0.10	2.4±0.14 2.3±0.12

FEV₁: Forced expiratory volume in 1 s, FVC: Forced vital capacity, PEFR: Postexpiratory flow rate, CFQ-R: Cystic fibrosis questionnaire revised, STAIC: State trait anxiety inventory, RCT: Randomized controlled trial

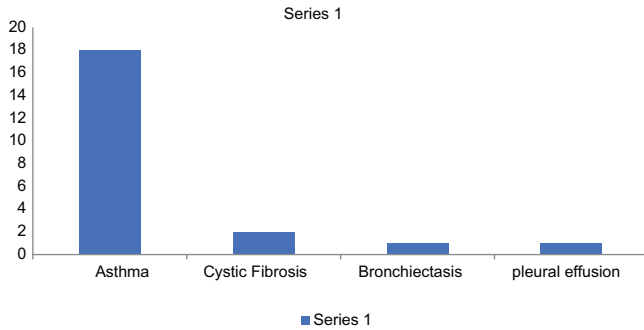


Figure 3: Figure 2 represents the number of studies found for different diseases. The maximum studies were found for asthma, followed by cystic fibrosis, bronchiectasis, and pleural effusion

actions of yoga among children with Bronchial asthma. Moreover, limited research has been conducted on children, yet this is the population who will be highly benefited as secondary changes are yet to happen. As mentioned above, limited work has been done on other respiratory disorders. There is considerable scope for further clinical research in these diseases, such as COPD and TB, among children and adolescents. Where no studies were identified. On TB and COPD, a plethora of evidence is available which has reported that yoga has beneficial results among adults that could be important to explore in LIMC countries, as the prevalence is still high. Moreover, by correcting the pathophysiology of the diseases through yoga among children, we can reduce the severity and chances of comorbidity of the diseases.

We encountered that most of the studies were conducted in India ($n = 13$), with a very limited number of studies in the US, Taiwan, and Egypt. The Indian population culturally accepts yoga which makes adherence better; however, in other countries, the adherence is not as high as in Indian studies. To add on, the Indian teachers are more trained in yoga than their counterparts and tend to provide a more integrative module rather than a single facet of yoga, such as meditation practice. Standardization in yoga training and intervention protocols across different regions poses a challenge to the global applicability and reproducibility of findings. These points highlight the need for further multicentric studies to assess the generalizability and effectiveness of yoga-based PR model in diverse cultural and healthcare contexts.

Yoga as a pulmonary rehabilitation tool

Yoga and PR are based on similar principles that are complementary to each other; however, there are some important differences that highlight the need for the formation of Yoga-based PR model. The major components of PR are exercise training, nutritional and psychological counseling, and optimal pharmacotherapy. In resource-constrained settings such as low- and middle-income countries, the implementation of conventional PR is often limited by the need for trained

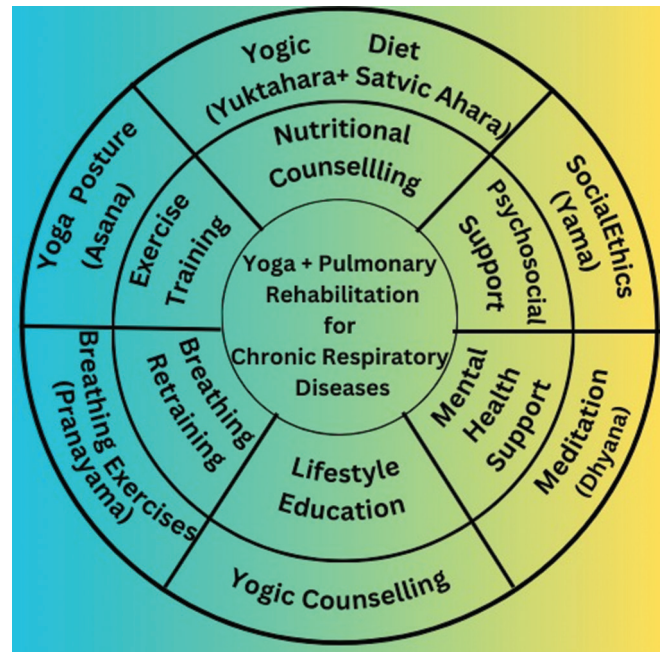


Figure 4: Represents yoga as a pulmonary rehabilitation tool

personnel and specialized infrastructure. In this context, yoga represents a culturally appropriate and logistically feasible adjunct. Its practice requires minimal equipment and can be conducted in nonclinical settings, making it an accessible complement to traditional PR strategies. However, direct comparative evidence is needed to clarify its role and inform integration into formal rehabilitation programs. Yoga, with its adaptable and holistic framework, holds potential as a complementary approach to conventional PR, particularly in resource-limited settings. Components of PR, such as breath control and exercise training, may be supported by yoga practices such as asanas and pranayama, while yogic counseling and dietary guidance may contribute to behavioral change and psychological well-being, and smoking cessation. A structured yoga-based model could be delivered by a small team of yoga-trained healthcare providers, including home-based or telemedicine platforms, requiring minimal space and equipment-features especially valuable in low-resource settings. Yogic breathing techniques such as Bibhagiya pranayama, Ujjayi Pranayama, and Nadi Shodhan share common objectives with PR breathing exercises like diaphragmatic breathing, including strengthening respiratory muscles and alleviating breathlessness. In addition, whole PR traditionally incorporates psychological counselling and behavioral change strategies to address stress and anxiety, and Yoga inherently integrates mindfulness, meditation, and self-regulatory practices that support emotional resilience and coping with chronic illness [Figure 4].

Although many studies have implemented yoga-based interventions within the PR framework, key components such as Yama and Niyama, yogic diet (dietary pattern recommended in traditional yogic texts emphasizing sattvic

foods), and yogic counseling (traditional form of guidance rooted in yogic philosophy) still remain underutilized. Incorporating these unexplored facets may enrich the comprehensiveness of integrative rehabilitation models. However, further high-quality comparative studies are needed to establish the efficacy, feasibility, and long-term outcomes of yoga as an adjunctive strategy within conventional PR protocols.

Conclusion

The result of our review suggests that, though there is a beneficial effect of yoga in respiratory diseases in the pediatric population, the generalizability is still difficult. There is scope for clinical trials in a spectrum of diseases. We have proposed a yoga-based PR module which could be beneficial for this vulnerable population. This model has the potential to improve patient outcomes by addressing not only physical rehabilitation but also psychological and emotional well-being.

Author contributions

Prof. Meenu Singh provided overall supervision, project administration, and critical revisions to the manuscript. Dr. Srijoy Mohanty conceptualized the study, supervised the research process, and guided methodological decisions. Monica Mahendru and Dheeraj Dabas reviewed, wrote, and edited the draft.

Ethical statement

Ethical statement was not required as it was a review; however, the study was registered in OSF (Open Science Framework – For scoping Reviews).

Acknowledgements

The authors express their appreciation to the department of telemedicine and department of Ayush for their help and guidance in this study. Special thanks to PhD scholar Nemichand Verma and research assistant Jorani Debbarma for their valuable insights and feedback. Finally, I acknowledge the unwavering support of all the other colleagues, whose encouragement has been instrumental in completing this work.

Financial support and sponsorship

Nil.

Conflicts of interest

There are no conflicts of interest.

References

- Momtazmanesh S, Moghaddam SS, Ghamari SH, Rad EM, Rezaei N, Shobeiri P, *et al.* Global burden of chronic respiratory diseases and risk factors, 1990–2019: An update from the Global Burden of Disease Study 2019. *EClinicalMedicine* 2023;59:101936.
- Pearce N, Ait-Khaled N, Beasley R, Mallol J, Keil U, Mitchell E, *et al.* ISAAC Phase Three Study Group. Asthma, Worldwide trends in the prevalence of asthma symptoms: phase III of the International Study of Asthma and Allergies in Childhood (ISAAC). Wellington, New Zealand.
- Syamlal G, Bhattacharya A, Dodd KE. Medical expenditures attributed to asthma and chronic obstructive pulmonary disease among workers – United States, 2011–2015. *Morb Mortal Wkly Rep* 2020;69:809–14.
- Barne M, Singh S, Mangal DK, Singh M, Awasthi S, Mahesh PA, *et al.* Global asthma network phase I, India: Results for allergic rhinitis and eczema in 127,309 children and adults. *J Allergy Clin Immunol Glob* 2022;1:51–60.
- Lai CK, Beasley R, Crane J, Foliaki S, Shah J, Weiland S, *et al.* Global variation in the prevalence and severity of asthma symptoms: Phase three of the international study of asthma and allergies in childhood (ISAAC). *Thorax* 2009;64:476–83.
- Shastri SS, Kabra M, Kabra SK, Pandey RM, Menon PS. Characterisation of mutations and genotype-phenotype correlation in cystic fibrosis: Experience from India. *J Cyst Fibros* 2008;7:110–5.
- Sharma S, Sarin R, Sahu G, Shukla G. Demographic profile, clinical and microbiological predictors of mortality amongst admitted pediatric TB patients in a tertiary referral tuberculosis hospital. *Indian J Tuberc* 2020;67:312–9.
- Agusti A, Celli BR, Criner GJ, Halpin D, Anzueto A, Barnes P, *et al.* Global initiative for chronic obstructive lung disease 2023 report: GOLD executive summary. *Eur Respir J* 2023;61:2300239.
- Rochester CL, Spruit MA, Holland AE. Pulmonary rehabilitation in 2021. *JAMA* 2021;326:969–70.
- Babita, Gulati K, Ray A. A clinical study to evaluate the effects of adjunct yogic intervention on pulmonary functions in patients of bronchial asthma. *J Clin Invest Stud* 2018;1:1–4.
- Fulambarker A, Farooki B, Kheir F, Copur AS, Srinivasan L, Schultz S. Effect of yoga in chronic obstructive pulmonary disease. *Am J Ther* 2012;19:96–100.
- Das RR, Sankar J, Kabra SK. Role of breathing exercises and yoga/pranayama in childhood asthma: A systematic review. *Curr Pediatr Rev* 2019;15:175–83.
- Tricco AC, Lillie E, Zarin W, O'Brien KK, Colquhoun H, Levac D, *et al.* PRISMA extension for scoping reviews (PRISMA-ScR): Checklist and explanation. *Ann Intern Med* 2018;169:467–73.
- Singh S, Yadav R. Bronchial asthma: A global health problem. *Int J Pharm Bio Sci* 2016;7:163–73.
- Singh S, Soni R, Singh KP, Tandon OP. Effect of yoga practices on pulmonary function tests including transfer factor of lung for carbon monoxide (TLCO) in asthma patients. *Indian J Physiol Pharmacol* 2012;56:63–8.
- Yadav P, Jain PK, Sharma BS, Sharma M. Yoga therapy as an adjuvant in management of asthma. *Indian J Pediatr* 2021;88:1127–34.
- Yadav R, Kabra SK, Yadav RK, Nandy A, Upadhyay AD, Ram Jat K, *et al.* Efficacy of Bhramari pranayama and Om chanting on asthma control, quality of life, and airway inflammation in asthmatic children: An open-label randomized controlled trial. *J Asthma* 2024;61:249–59.
- Freitas DA, Holloway EA, Bruno SS, Chaves GS, Fregonezi GA, Mendonça KP. Breathing exercises for adults with asthma. *Cochrane Database Syst Rev* 2013;p. CD001277.
- Jasrotia R, Mondal S, Kumar V, Gandhi A. Impact of adjunct treatment with yoga on severity, illness score, and drug dosage in controlled asthmatic children. *Natl J Physiol Pharm Pharmacol* 2019;9:1.

20. Nasrollahpor S, Bahrami A, Aliakbari M, Haghani S, Sabeti F. The effect of yoga breathing exercises on quality of life in children with asthma. *Jundishapur J Chronic Dis Care* 2024;13:e138819.
21. Pandya S. Yoga education program for reducing drug dependency and promoting better asthma control for chronic asthmatic children: A multicity experiment. *Glob Pediatr Health* 2019;6:2333794X19837455.
22. Sodhi C, Singh S, Bery A. Assessment of the quality of life in patients with bronchial asthma, before and after yoga: A randomised trial. *Iran J Allergy Asthma Immunol* 2014;13:55-60.
23. Khanam AA, Sachdeva U, Guleria R, Deepak KK. Study of pulmonary and autonomic functions of asthma patients after yoga training. *Indian J Physiol Pharmacol* 1996;40:318-24.
24. Ruddy J, Emerson J, McNamara S, Genatossio A, Breuner C, Weber T, *et al.* Yoga as a therapy for adolescents and young adults with cystic fibrosis: A pilot study. *Glob Adv Health Med* 2015;4:32-6.
25. McNamara C, Johnson M, Read L, Vander Velden H, Thygeson M, Liu M, *et al.* Yoga therapy in children with cystic fibrosis decreases immediate anxiety and joint pain. *Evid Based Complement Alternat Med* 2016;2016:9429504.
26. Medvedev IN, Kachenkova ES, Zbrueva YV, Nalobina AN, Kulkova IV. Physiological beneficial changes in the respiratory system in adolescents with chronic obstructive bronchitis as a result of the use of a set of yogic exercises. *Biomed Pharmacol J* 2021;14:1323-9.
27. Prakasamma M, Bhaduri A. A study of yoga as a nursing intervention in the care of patients with pleural effusion. *J Adv Nurs* 1984;9:127-33.

Supplementary Table 1: Search strategies

Serial number	Data base	Search strategy	Result
1	PubMed (Yoga)	“yoga”[MeSH Terms] OR “yoga”[All Fields] OR “meditate”[All Fields] OR “meditated”[All Fields] OR “meditating”[All Fields] OR “meditation”[MeSH Terms] OR “meditation”[All Fields] OR “meditations”[All Fields] OR “meditation s”[All Fields] OR “meditational”[All Fields] OR “meditative”[All Fields] OR “meditator”[All Fields] OR “meditators”[All Fields] OR “asana”[All Fields] OR “asanas”[All Fields] OR “pranayama”[All Fields] OR “pranayamas”[All Fields] OR “mindful”[All Fields] OR “mindfulness”[MeSH Terms] OR “mindfulness”[All Fields]	40,565
2	Pediatrics	“paediatrics”[All Fields] OR “pediatrics”[MeSH Terms] OR “pediatrics”[All Fields] OR “paediatric”[All Fields] OR “pediatric”[All Fields] OR (“child”[MeSH Terms] OR “child”[All Fields] OR “children”[All Fields] OR “child s”[All Fields] OR “children s”[All Fields] OR “childrens”[All Fields] OR “childs”[All Fields]) OR (“adolescences”[All Fields] OR “adolescence”[All Fields] OR “adolescent”[MeSH Terms] OR “adolescent”[All Fields] OR “adolescent s”[All Fields] OR “adolescence”[All Fields] OR “adolescents”[All Fields] OR “adolescent s”[All Fields]) OR (“adolescent”[MeSH Terms] OR “adolescent”[All Fields] OR “youth”[All Fields] OR “youths”[All Fields] OR “youth s”[All Fields]) OR (“adolescent”[MeSH Terms] OR “adolescent”[All Fields] OR “teenage”[All Fields] OR “teenager”[All Fields] OR “teenagers”[All Fields] OR “teenaged”[All Fields] OR “teenager s”[All Fields] OR “teenages”[All Fields]) OR (“child, preschool”[MeSH Terms] OR (“child”[All Fields] AND “preschool”[All Fields]) OR “preschool child”[All Fields] OR (“preschool”[All Fields] AND “child”[All Fields]))	4,746,349
3	Asthma	“asthma”[MeSH Terms] OR “asthma”[All Fields] OR “asthmas”[All Fields] OR “asthma s”[All Fields] OR (“bronchoconstricted”[All Fields] OR “bronchoconstricting”[All Fields] OR “bronchoconstriction”[MeSH Terms] OR “bronchoconstriction”[All Fields] OR “bronchoconstrictions”[All Fields] OR “bronchoconstrictive”[All Fields]) OR (“respiratory tract infections”[MeSH Terms] OR (“respiratory”[All Fields] AND “tract”[All Fields] AND “infections”[All Fields]) OR “respiratory tract infections”[All Fields]) OR (“respiratory sounds”[MeSH Terms] OR (“respiratory”[All Fields] AND “sounds”[All Fields]) OR “respiratory sounds”[All Fields] OR “wheeze”[All Fields] OR “wheezes”[All Fields] OR “wheezing”[All Fields] OR “wheezed”[All Fields]) OR “bronchial asthma”	871,363
4		“tuberculosis, pulmonary”[MeSH Terms] OR (“tuberculosis”[All Fields] AND “pulmonary”[All Fields]) OR “pulmonary tuberculosis”[All Fields] OR (“pulmonary”[All Fields] AND “tuberculosis”[All Fields]) OR (“tuberculosi”[All Fields] OR “tuberculosis”[MeSH Terms] OR “tuberculosis”[All Fields] OR “tuberculoses”[All Fields] OR “tuberculosis s”[All Fields]) OR (“paediatrics”[All Fields] OR “pediatrics”[MeSH Terms] OR “pediatrics”[All Fields] OR “paediatric”[All Fields] OR “pediatric”[All Fields]) AND (“tuberculosi”[All Fields] OR “tuberculosis”[MeSH Terms] OR “tuberculosis”[All Fields] OR “tuberculoses”[All Fields] OR “tuberculosis s”[All Fields]) OR (“tuberculosi”[All Fields] OR “tuberculosis”[MeSH Terms] OR “tuberculosis”[All Fields] OR “tuberculoses”[All Fields] OR “tuberculosis s”[All Fields]) AND (“bacteraemia”[All Fields] OR “bacteremia”[MeSH Terms] OR “bacteremia”[All Fields] OR “bacteraemias”[All Fields] OR “bacteremias”[All Fields]) OR (“blood circulation”[MeSH Terms] OR (“blood”[All Fields] AND “circulation”[All Fields]) OR “blood circulation”[All Fields] OR “bloodstream”[All Fields] OR “bloodstreams”[All Fields]) AND (“tuberculosi”[All Fields] OR “tuberculosis”[MeSH Terms] OR “tuberculosis”[All Fields] OR “tuberculoses”[All Fields] OR “tuberculosis s”[All Fields]))	871,363
5		“cystic fibrosis”[MeSH Terms] OR (“cystic”[All Fields] AND “fibrosis”[All Fields]) OR “cystic fibrosis”[All Fields] OR (“cystic fibrosis”[MeSH Terms] OR (“cystic”[All Fields] AND “fibrosis”[All Fields]) OR “cystic fibrosis”[All Fields] OR “mucoviscidosis”[All Fields])	62,563
6		“pulmonary disease, chronic obstructive”[MeSH Terms] OR (“pulmonary”[All Fields] AND “disease”[All Fields] AND “chronic”[All Fields] AND “obstructive”[All Fields]) OR “chronic obstructive pulmonary disease”[All Fields] OR (“chronic”[All Fields] AND “obstructive”[All Fields] AND “pulmonary”[All Fields] AND “disease”[All Fields]) OR (“lung diseases, obstructive”[MeSH Terms] OR (“lung”[All Fields] AND “diseases”[All Fields] AND “obstructive”[All Fields]) OR “obstructive lung diseases”[All Fields] OR (“obstructive”[All Fields] AND “airway”[All Fields] AND “disease”[All Fields]) OR “obstructive airway disease”[All Fields]) OR (“airflow”[All Fields] OR “airflows”[All Fields]) AND (“obstruct”[All Fields] OR “obstructed”[All Fields] OR “obstructing”[All Fields] OR “obstruction”[All Fields] OR “obstructions”[All Fields] OR “obstructive”[All Fields] OR “obstructs”[All Fields])) OR (“bronchitis, chronic”[MeSH Terms] OR (“bronchitis”[All Fields] AND “chronic”[All Fields]) OR	295,339

Contd...

Supplementary Table 1: Contd...

Serial number	Data base	Search strategy	Result
7		“chronic bronchitis”[All Fields] OR (“chronic”[All Fields] AND “bronchitis”[All Fields]) OR (“emphysema”[MeSH Terms] OR “emphysema”[All Fields] OR “emphysemas”[All Fields] OR “pulmonary emphysema”[MeSH Terms] OR (“pulmonary”[All Fields] AND “emphysema”[All Fields]) OR “pulmonary emphysema”[All Fields]) “rhinitis, allergic”[MeSH Terms] OR (“rhinitis”[All Fields] AND “allergic”[All Fields]) OR “allergic rhinitis”[All Fields] OR (“allergic”[All Fields] AND “rhinitis”[All Fields]) OR (“rhinitis, allergic, seasonal”[MeSH Terms] OR (“rhinitis”[All Fields] AND “allergic”[All Fields] AND “seasonal”[All Fields]) OR “seasonal allergic rhinitis”[All Fields] OR (“hay”[All Fields] AND “fever”[All Fields]) OR “hay fever”[All Fields]) OR (“perennial”[All Fields] OR “perennialism”[All Fields] OR “perenniality”[All Fields] OR “perennially”[All Fields] OR “perennials”[All Fields] OR “perennis”[All Fields]) AND (“rhinitis”[MeSH Terms] OR “rhinitis”[All Fields] OR “rhinitides”[All Fields]) OR (“chronic”[All Fields] OR “chronical”[All Fields] OR “chronically”[All Fields] OR “chronicities”[All Fields] OR “chronicity”[All Fields] OR “chronicization”[All Fields] OR “chronics”[All Fields]) AND (“rhinitis”[MeSH Terms] OR “rhinitis”[All Fields] OR “rhinitides”[All Fields])	44,386
8		((“pneumonia”[MeSH Terms] OR “pneumonia”[All Fields] OR “pneumonias”[All Fields] OR “pneumoniae”[All Fields] OR “pneumoniae s”[All Fields] OR (“communal”[All Fields] OR “communalism”[All Fields] OR “communalities”[All Fields] OR “communality”[All Fields] OR “communally”[All Fields] OR “commune”[All Fields] OR “communes”[All Fields] OR “community s”[All Fields] OR “communities”[All Fields] OR “residence characteristics”[MeSH Terms] OR (“residence”[All Fields] AND “characteristics”[All Fields]) OR “residence characteristics”[All Fields] OR “communities”[All Fields] OR “community”[All Fields]) AND “acquired”[All Fields] AND (“pneumonia”[MeSH Terms] OR “pneumonia”[All Fields] OR “pneumonias”[All Fields] OR “pneumoniae”[All Fields] OR “pneumoniae s”[All Fields])) OR (“sever”[All Fields] OR “severe”[All Fields] OR “severed”[All Fields] OR “severely”[All Fields] OR “severer”[All Fields] OR “severes”[All Fields] OR “severing”[All Fields] OR “severities”[All Fields] OR “severity”[All Fields] OR “severs”[All Fields])) OR (“pleural effusion”[MeSH Terms] OR (“pleural”[All Fields] AND “effusion”[All Fields]) OR “pleural effusion”[All Fields]) OR (“pneumonia, necrotizing”[MeSH Terms] OR (“pneumonia”[All Fields] AND “necrotizing”[All Fields]) OR “necrotizing pneumonia”[All Fields] OR (“necrotizing”[All Fields] AND “pneumonia”[All Fields])) AND ((“laboratorial”[All Fields] OR “laboratories”[MeSH Terms] OR “laboratories”[All Fields] OR “laboratory”[All Fields] OR “laboratory s”[All Fields]) AND (“factor”[All Fields] OR “factor s”[All Fields] OR “factors”[All Fields]))	31,412
1+2		(“yoga”[MeSH Terms] OR “yoga”[All Fields] OR “meditate”[All Fields] OR “meditated”[All Fields] OR “meditating”[All Fields] OR “meditation”[MeSH Terms] OR “meditation”[All Fields] OR “meditations”[All Fields] OR “meditation s”[All Fields] OR “meditational”[All Fields] OR “meditative”[All Fields] OR “meditator”[All Fields] OR “meditators”[All Fields] OR “asana”[All Fields] OR “asanas”[All Fields] OR “pranayama”[All Fields] OR “pranayamas”[All Fields] OR “mindful”[All Fields] OR “mindfulness”[MeSH Terms] OR “mindfulness”[All Fields]) AND (“paediatrics”[All Fields] OR “pediatrics”[MeSH Terms] OR “pediatrics”[All Fields] OR “paediatric”[All Fields] OR “pediatric”[All Fields] OR (“child”[MeSH Terms] OR “child”[All Fields] OR “children”[All Fields] OR “child s”[All Fields] OR “children s”[All Fields] OR “childrens”[All Fields] OR “childs”[All Fields] OR (“adolescences”[All Fields] OR “adolescence”[All Fields] OR “adolescent”[MeSH Terms] OR “adolescent”[All Fields] OR “adolescence”[All Fields] OR “adolescents”[All Fields] OR “adolescent s”[All Fields]) OR (“adolescent”[MeSH Terms] OR “adolescent”[All Fields] OR “youth”[All Fields] OR “youths”[All Fields] OR “youth s”[All Fields]) OR (“adolescent”[MeSH Terms] OR “adolescent”[All Fields] OR “teenage”[All Fields] OR “teenager”[All Fields] OR “teenagers”[All Fields] OR “teenaged”[All Fields] OR “teenager s”[All Fields] OR “teenages”[All Fields]) OR (“child, preschool”[MeSH Terms] OR (“child”[All Fields] AND “preschool”[All Fields]) OR “preschool child”[All Fields] OR (“preschool”[All Fields] AND “child”[All Fields]))	5621
1+3		(“yoga”[MeSH Terms] OR “yoga”[All Fields] OR “meditate”[All Fields] OR “meditated”[All Fields] OR “meditating”[All Fields] OR “meditation”[MeSH Terms] OR “meditation”[All Fields] OR “meditations”[All Fields] OR “meditation s”[All Fields] OR “meditational”[All Fields] OR “meditative”[All Fields] OR “meditator”[All Fields] OR “meditators”[All Fields] OR “asana”[All Fields] OR “asanas”[All Fields] OR “pranayama”[All Fields] OR “pranayamas”[All Fields] OR “mindful”[All Fields] OR “mindfulness”[MeSH Terms] OR “mindfulness”[All Fields]) AND (“asthma”[MeSH Terms] OR “asthma”[All Fields] OR “asthmas”[All Fields] OR asthma	1072

Contd...

Supplementary Table 1: Contd...

Serial number	Data base	Search strategy	Result
1+4		“s”[All Fields] OR (“bronchoconstricted”[All Fields] OR “bronchoconstricting”[All Fields] OR “bronchoconstriction”[MeSH Terms] OR “bronchoconstriction”[All Fields] OR “bronchoconstrictions”[All Fields] OR “bronchoconstrictive”[All Fields]) OR (“respiratory tract infections”[MeSH Terms] OR (“respiratory”[All Fields] AND “tract”[All Fields] AND “infections”[All Fields]) OR “respiratory tract infections”[All Fields]) OR (“respiratory sounds”[MeSH Terms] OR (“respiratory”[All Fields] AND “sounds”[All Fields]) OR “respiratory sounds”[All Fields] OR “wheeze”[All Fields] OR “wheezes”[All Fields] OR “wheezing”[All Fields] OR “wheezed”[All Fields]) OR “bronchial asthma”) (“yoga”[MeSH Terms] OR “yoga”[All Fields] OR “meditate”[All Fields] OR “meditated”[All Fields] OR “meditating”[All Fields] OR “meditation”[MeSH Terms] OR “meditation”[All Fields] OR “meditations”[All Fields] OR “meditation s”[All Fields] OR “meditational”[All Fields] OR “meditative”[All Fields] OR “meditator”[All Fields] OR “meditators”[All Fields] OR “asana”[All Fields] OR “asanas”[All Fields] OR “pranayama”[All Fields] OR “pranayamas”[All Fields] OR “mindful”[All Fields] OR “mindfulness”[MeSH Terms] OR “mindfulness”[All Fields]) AND (“tuberculosis, pulmonary”[MeSH Terms] OR (“tuberculosis”[All Fields] AND “pulmonary”[All Fields]) OR “pulmonary tuberculosis”[All Fields] OR (“pulmonary”[All Fields] AND “tuberculosis”[All Fields]) OR (“tuberculosi”[All Fields] OR “tuberculosis”[MeSH Terms] OR “tuberculosis”[All Fields] OR “tuberculoses”[All Fields] OR “tuberculosis s”[All Fields]) OR (“paediatrics”[All Fields] OR “pediatrics”[MeSH Terms] OR “pediatrics”[All Fields] OR “paediatric”[All Fields] OR “pediatric”[All Fields]) AND (“tuberculosi”[All Fields] OR “tuberculosis”[MeSH Terms] OR “tuberculosis”[All Fields] OR “tuberculoses”[All Fields] OR “tuberculosis s”[All Fields] OR “tuberculosis”[All Fields]) OR (“tuberculosi”[All Fields] OR “tuberculosis”[MeSH Terms] OR “tuberculosis”[All Fields] OR “tuberculoses”[All Fields] OR “tuberculosis s”[All Fields]) AND (“bacteraemia”[All Fields] OR “bacteremia”[MeSH Terms] OR “bacteremia”[All Fields] OR “bacteraemias”[All Fields] OR “bacteremias”[All Fields])) OR (“blood circulation”[MeSH Terms] OR (“blood”[All Fields] AND “circulation”[All Fields]) OR “blood circulation”[All Fields] OR “bloodstream”[All Fields] OR “bloodstreams”[All Fields]) AND (“tuberculosi”[All Fields] OR “tuberculosis”[MeSH Terms] OR “tuberculosis”[All Fields] OR “tuberculoses”[All Fields] OR “tuberculosis s”[All Fields]))	44
1+5		(“yoga”[MeSH Terms] OR “yoga”[All Fields] OR “meditate”[All Fields] OR “meditated”[All Fields] OR “meditating”[All Fields] OR “meditation”[MeSH Terms] OR “meditation”[All Fields] OR “meditations”[All Fields] OR “meditation s”[All Fields] OR “meditational”[All Fields] OR “meditative”[All Fields] OR “meditator”[All Fields] OR “meditators”[All Fields] OR “asana”[All Fields] OR “asanas”[All Fields] OR “pranayama”[All Fields] OR “pranayamas”[All Fields] OR “mindful”[All Fields] OR “mindfulness”[MeSH Terms] OR “mindfulness”[All Fields]) AND (“cystic fibrosis”[MeSH Terms] OR (“cystic”[All Fields] AND “fibrosis”[All Fields]) OR “cystic fibrosis”[All Fields] OR (“cystic fibrosis”[MeSH Terms] OR (“cystic”[All Fields] AND “fibrosis”[All Fields]) OR “cystic fibrosis”[All Fields] OR “mucoviscidosis”[All Fields]))	29
1+6		H Terms] OR “meditation”[All Fields] OR “meditations”[All Fields] OR “meditation s”[All Fields] OR “meditational”[All Fields] OR “meditative”[All Fields] OR “meditator”[All Fields] OR “meditators”[All Fields] OR “asana”[All Fields] OR “asanas”[All Fields] OR “pranayama”[All Fields] OR “pranayamas”[All Fields] OR “mindful”[All Fields] OR “mindfulness”[MeSH Terms] OR “mindfulness”[All Fields]) AND (“pulmonary disease, chronic obstructive”[MeSH Terms] OR (“pulmonary”[All Fields] AND “disease”[All Fields] AND “chronic”[All Fields] AND “obstructive”[All Fields]) OR “chronic obstructive pulmonary disease”[All Fields] OR (“chronic”[All Fields] AND “obstructive”[All Fields] AND “pulmonary”[All Fields] AND “disease”[All Fields]) OR (“lung diseases, obstructive”[MeSH Terms] OR (“lung”[All Fields] AND “diseases”[All Fields] AND “obstructive”[All Fields]) OR “obstructive lung diseases”[All Fields] OR (“obstructive”[All Fields] AND “airway”[All Fields] AND “disease”[All Fields]) OR “obstructive airway disease”[All Fields]) OR (“airflow”[All Fields] OR “airflows”[All Fields]) AND (“obstruct”[All Fields] OR “obstructed”[All Fields] OR “obstructing”[All Fields] OR “obstruction”[All Fields] OR “obstructions”[All Fields] OR “obstructive”[All Fields] OR “obstructs”[All Fields])) OR (“bronchitis, chronic”[MeSH Terms] OR (“bronchitis”[All Fields] AND “chronic”[All Fields]) OR “chronic bronchitis”[All Fields] OR (“chronic”[All Fields] AND “bronchitis”[All Fields])) OR (“emphysema”[MeSH Terms] OR “emphysema”[All Fields] OR “emphysemas”[All Fields] OR “pulmonary emphysema”[MeSH Terms] OR (“pulmonary”[All Fields] AND “emphysema”[All Fields]) OR “pulmonary emphysema”[All Fields]))	299

Contd...

Supplementary Table 1: Contd...

Serial number	Data base	Search strategy	Result
1+7		(“yoga”[MeSH Terms] OR “yoga”[All Fields] OR “meditate”[All Fields] OR “meditated”[All Fields] OR “meditating”[All Fields] OR “meditation”[MeSH Terms] OR “meditation”[All Fields] OR “meditations”[All Fields] OR “meditation s”[All Fields] OR “meditational”[All Fields] OR “meditative”[All Fields] OR “meditator”[All Fields] OR “meditators”[All Fields] OR “asana”[All Fields] OR “asanas”[All Fields] OR “pranayama”[All Fields] OR “pranayamas”[All Fields] OR “mindful”[All Fields] OR “mindfulness”[MeSH Terms] OR “mindfulness”[All Fields]) AND (“rhinitis, allergic”[MeSH Terms] OR (“rhinitis”[All Fields] AND “allergic”[All Fields]) OR “allergic rhinitis”[All Fields] OR (“allergic”[All Fields] AND “rhinitis”[All Fields]) OR (“rhinitis, allergic, seasonal”[MeSH Terms] OR (“rhinitis”[All Fields] AND “allergic”[All Fields] AND “seasonal”[All Fields]) OR “seasonal allergic rhinitis”[All Fields] OR (“hay”[All Fields] AND “fever”[All Fields]) OR “hay fever”[All Fields]) OR (“perennial”[All Fields] OR “perennialism”[All Fields] OR “perenniality”[All Fields] OR “perennially”[All Fields] OR “perennials”[All Fields] OR “perennis”[All Fields]) AND (“rhinitis”[MeSH Terms] OR “rhinitis”[All Fields] OR “rhinitides”[All Fields])) OR (“chronic”[All Fields] OR “chronical”[All Fields] OR “chronically”[All Fields] OR “chronicities”[All Fields] OR “chronicity”[All Fields] OR “chronicization”[All Fields] OR “chronics”[All Fields]) AND (“rhinitis”[MeSH Terms] OR “rhinitis”[All Fields] OR “rhinitides”[All Fields]))	21
1+8		(“yoga”[MeSH Terms] OR “yoga”[All Fields] OR “meditate”[All Fields] OR “meditated”[All Fields] OR “meditating”[All Fields] OR “meditation”[MeSH Terms] OR “meditation”[All Fields] OR “meditations”[All Fields] OR “meditation s”[All Fields] OR “meditational”[All Fields] OR “meditative”[All Fields] OR “meditator”[All Fields] OR “meditators”[All Fields] OR “asana”[All Fields] OR “asanas”[All Fields] OR “pranayama”[All Fields] OR “pranayamas”[All Fields] OR “mindful”[All Fields] OR “mindfulness”[MeSH Terms] OR “mindfulness”[All Fields]) AND (((“pneumonia”[MeSH Terms] OR “pneumonia”[All Fields] OR “pneumonias”[All Fields] OR “pneumoniae”[All Fields] OR “pneumoniae s”[All Fields] OR (“communal”[All Fields] OR “communalism”[All Fields] OR “communalities”[All Fields] OR “communality”[All Fields] OR “communally”[All Fields] OR “commune”[All Fields] OR “communes”[All Fields] OR “community s”[All Fields] OR “communities”[All Fields] OR “residence characteristics”[MeSH Terms] OR (“residence”[All Fields] AND “characteristics”[All Fields]) OR “residence characteristics”[All Fields] OR “communities”[All Fields] OR “community”[All Fields]) AND “acquired”[All Fields] AND (“pneumonia”[MeSH Terms] OR “pneumonia”[All Fields] OR “pneumonias”[All Fields] OR “pneumoniae”[All Fields] OR “pneumoniae s”[All Fields])) OR (“sever”[All Fields] OR “severe”[All Fields] OR “severed”[All Fields] OR “severely”[All Fields] OR “severer”[All Fields] OR “severes”[All Fields] OR “severing”[All Fields] OR “severities”[All Fields] OR “severity”[All Fields] OR “severs”[All Fields])) OR (“pleural effusion”[MeSH Terms] OR (“pleural”[All Fields] AND “effusion”[All Fields]) OR “pleural effusion”[All Fields]) OR (“pneumonia, necrotizing”[MeSH Terms] OR (“pneumonia”[All Fields] AND “necrotizing”[All Fields]) OR “necrotizing pneumonia”[All Fields] OR (“necrotizing”[All Fields] AND “pneumonia”[All Fields])) AND (“laboratorial”[All Fields] OR “laboratories”[MeSH Terms] OR “laboratories”[All Fields] OR “laboratory”[All Fields] OR “laboratory s”[All Fields]) AND (“factor”[All Fields] OR “factor s”[All Fields] OR “factors”[All Fields]))	62
1+2+3		((“yoga”[MeSH Terms] OR “yoga”[All Fields] OR “meditate”[All Fields] OR “meditated”[All Fields] OR “meditating”[All Fields] OR “meditation”[MeSH Terms] OR “meditation”[All Fields] OR “meditations”[All Fields] OR “meditation s”[All Fields] OR “meditational”[All Fields] OR “meditative”[All Fields] OR “meditator”[All Fields] OR “meditators”[All Fields] OR “asana”[All Fields] OR “asanas”[All Fields] OR “pranayama”[All Fields] OR “pranayamas”[All Fields] OR “mindful”[All Fields] OR “mindfulness”[MeSH Terms] OR “mindfulness”[All Fields]) AND (“paediatrics”[All Fields] OR “pediatrics”[MeSH Terms] OR “pediatrics”[All Fields] OR “paediatric”[All Fields] OR “pediatric”[All Fields] OR (“child”[MeSH Terms] OR “child”[All Fields] OR “children”[All Fields] OR “child s”[All Fields] OR “children s”[All Fields] OR “childrens”[All Fields] OR “childs”[All Fields]) OR (“adolescences”[All Fields] OR “adolescence”[All Fields] OR “adolescent”[MeSH Terms] OR “adolescent”[All Fields] OR “adolescence”[All Fields] OR “adolescents”[All Fields] OR “adolescent s”[All Fields]) OR (“adolescent”[MeSH Terms] OR “adolescent”[All Fields] OR “youth”[All Fields] OR “youths”[All Fields] OR “youth s”[All Fields]) OR (“adolescent”[MeSH Terms] OR “adolescent”[All Fields] OR “teenage”[All Fields] OR “teenager”[All Fields] OR “teenagers”[All Fields] OR “teenaged”[All Fields] OR “teenager s”[All Fields] OR “teenages”[All Fields]) OR (“child, preschool”[MeSH Terms] OR (“child”[All Fields] AND “preschool”[All Fields]) OR “preschool	248

Contd...

Supplementary Table 1: Contd...

Serial number	Data base	Search strategy	Result
1+2+4		child"[All Fields] OR ("preschool"[All Fields] AND "child"[All Fields])))) AND ("asthma"[MeSH Terms] OR "asthma"[All Fields] OR "asthmas"[All Fields] OR "asthma s"[All Fields] OR ("bronchoconstricted"[All Fields] OR "bronchoconstricting"[All Fields] OR "bronchoconstriction"[MeSH Terms] OR "bronchoconstriction"[All Fields] OR "bronchoconstrictions"[All Fields] OR "bronchoconstrictive"[All Fields]) OR ("respiratory tract infections"[MeSH Terms] OR ("respiratory"[All Fields] AND "tract"[All Fields] AND "infections"[All Fields]) OR "respiratory tract infections"[All Fields]) OR ("respiratory sounds"[MeSH Terms] OR ("respiratory"[All Fields] AND "sounds"[All Fields]) OR "respiratory sounds"[All Fields] OR "wheeze"[All Fields] OR "wheezes"[All Fields] OR "wheezing"[All Fields] OR "wheezed"[All Fields] OR "bronchial asthma") ((("yoga"[MeSH Terms] OR "yoga"[All Fields] OR "meditate"[All Fields] OR "meditated"[All Fields] OR "meditating"[All Fields] OR "meditation"[MeSH Terms] OR "meditation"[All Fields] OR "meditations"[All Fields] OR "meditation s"[All Fields] OR "meditational"[All Fields] OR "meditative"[All Fields] OR "meditator"[All Fields] OR "meditators"[All Fields] OR "asana"[All Fields] OR "asanas"[All Fields] OR "pranayama"[All Fields] OR "pranayamas"[All Fields] OR "mindful"[All Fields] OR "mindfulness"[MeSH Terms] OR "mindfulness"[All Fields]) AND ("paediatrics"[All Fields] OR "pediatrics"[MeSH Terms] OR "pediatrics"[All Fields] OR "paediatric"[All Fields] OR "pediatric"[All Fields] OR ("child"[MeSH Terms] OR "child"[All Fields] OR "children"[All Fields] OR "child s"[All Fields] OR "children s"[All Fields] OR "childrens"[All Fields] OR "childs"[All Fields]) OR ("adolescences"[All Fields] OR "adolescence"[All Fields] OR "adolescent"[MeSH Terms] OR "adolescent"[All Fields] OR "adolescence"[All Fields] OR "adolescents"[All Fields] OR "adolescent s"[All Fields]) OR ("adolescent"[MeSH Terms] OR "adolescent"[All Fields] OR "youth"[All Fields] OR "youths"[All Fields] OR "youth s"[All Fields]) OR ("adolescent"[MeSH Terms] OR "adolescent"[All Fields] OR "teenage"[All Fields] OR "teenager"[All Fields] OR "teenagers"[All Fields] OR "teenaged"[All Fields] OR "teenager s"[All Fields] OR "teenages"[All Fields]) OR ("child, preschool"[MeSH Terms] OR ("child"[All Fields] AND "preschool"[All Fields]) OR "preschool child"[All Fields] OR ("preschool"[All Fields] AND "child"[All Fields])))) AND ("tuberculosis, pulmonary"[MeSH Terms] OR ("tuberculosis"[All Fields] AND "pulmonary"[All Fields]) OR "pulmonary tuberculosis"[All Fields] OR ("pulmonary"[All Fields] AND "tuberculosis"[All Fields]) OR ("tuberculosi"[All Fields] OR "tuberculosis"[MeSH Terms] OR "tuberculosis"[All Fields] OR "tuberculoses"[All Fields] OR "tuberculosis s"[All Fields]) OR ((("paediatrics"[All Fields] OR "pediatrics"[MeSH Terms] OR "pediatrics"[All Fields] OR "paediatric"[All Fields] OR "pediatric"[All Fields]) AND ("tuberculosi"[All Fields] OR "tuberculosis"[MeSH Terms] OR "tuberculosis"[All Fields] OR "tuberculoses"[All Fields] OR "tuberculosis s"[All Fields])) OR ((("tuberculosi"[All Fields] OR "tuberculosis"[MeSH Terms] OR "tuberculosis"[All Fields] OR "tuberculoses"[All Fields] OR "tuberculosis s"[All Fields]) AND ("bacteraemia"[All Fields] OR "bacteremia"[MeSH Terms] OR "bacteremia"[All Fields] OR "bacteraemias"[All Fields] OR "bacteremias"[All Fields])) OR ((("blood circulation"[MeSH Terms] OR ("blood"[All Fields] AND "circulation"[All Fields]) OR "blood circulation"[All Fields] OR "bloodstream"[All Fields] OR "bloodstreams"[All Fields]) AND ("tuberculosi"[All Fields] OR "tuberculosis"[MeSH Terms] OR "tuberculosis"[All Fields] OR "tuberculoses"[All Fields] OR "tuberculosis s"[All Fields]))	7
1+2+5		((("yoga"[MeSH Terms] OR "yoga"[All Fields] OR "meditate"[All Fields] OR "meditated"[All Fields] OR "meditating"[All Fields] OR "meditation"[MeSH Terms] OR "meditation"[All Fields] OR "meditations"[All Fields] OR "meditation s"[All Fields] OR "meditational"[All Fields] OR "meditative"[All Fields] OR "meditator"[All Fields] OR "meditators"[All Fields] OR "asana"[All Fields] OR "asanas"[All Fields] OR "pranayama"[All Fields] OR "pranayamas"[All Fields] OR "mindful"[All Fields] OR "mindfulness"[MeSH Terms] OR "mindfulness"[All Fields]) AND ("paediatrics"[All Fields] OR "pediatrics"[MeSH Terms] OR "pediatrics"[All Fields] OR "paediatric"[All Fields] OR "pediatric"[All Fields] OR ("child"[MeSH Terms] OR "child"[All Fields] OR "children"[All Fields] OR "child s"[All Fields] OR "children s"[All Fields] OR "childrens"[All Fields] OR "childs"[All Fields]) OR ("adolescences"[All Fields] OR "adolescence"[All Fields] OR "adolescent"[MeSH Terms] OR "adolescent"[All Fields] OR "adolescence"[All Fields] OR "adolescents"[All Fields] OR "adolescent s"[All Fields]) OR ("adolescent"[MeSH Terms] OR "adolescent"[All Fields] OR "youth"[All Fields] OR "youths"[All Fields] OR "youth s"[All Fields]) OR ("adolescent"[MeSH Terms] OR "adolescent"[All Fields] OR "teenage"[All Fields] OR "teenager"[All Fields] OR "teenagers"[All Fields] OR "teenaged"[All Fields] OR "teenager s"[All Fields] OR "teenages"[All Fields]) OR ("child,	14

Contd...

Supplementary Table 1: Contd...

Serial number	Data base	Search strategy	Result
1+2+6		preschool"[MeSH Terms] OR ("child"[All Fields] AND "preschool"[All Fields]) OR "preschool child"[All Fields] OR ("preschool"[All Fields] AND "child"[All Fields])) AND ("cystic fibrosis"[MeSH Terms] OR ("cystic"[All Fields] AND "fibrosis"[All Fields]) OR "cystic fibrosis"[All Fields] OR ("cystic fibrosis"[MeSH Terms] OR ("cystic"[All Fields] AND "fibrosis"[All Fields]) OR "cystic fibrosis"[All Fields] OR "mucoviscidosis"[All Fields])) ((("yoga"[MeSH Terms] OR "yoga"[All Fields] OR "meditate"[All Fields] OR "meditated"[All Fields] OR "meditating"[All Fields] OR "meditation"[MeSH Terms] OR "meditation"[All Fields] OR "meditations"[All Fields] OR "meditation s"[All Fields] OR "meditational"[All Fields] OR "meditative"[All Fields] OR "meditator"[All Fields] OR "meditators"[All Fields] OR "asana"[All Fields] OR "asanas"[All Fields] OR "pranayama"[All Fields] OR "pranayamas"[All Fields] OR "mindful"[All Fields] OR "mindfulness"[MeSH Terms] OR "mindfulness"[All Fields]) AND ("paediatrics"[All Fields] OR "pediatrics"[MeSH Terms] OR "pediatrics"[All Fields] OR "paediatric"[All Fields] OR "pediatric"[All Fields] OR ("child"[MeSH Terms] OR "child"[All Fields] OR "children"[All Fields] OR "child s"[All Fields] OR "children s"[All Fields] OR "childrens"[All Fields] OR "childs"[All Fields]) OR ("adolescences"[All Fields] OR "adolescence"[All Fields] OR "adolescent"[MeSH Terms] OR "adolescent"[All Fields] OR "adolescence"[All Fields] OR "adolescents"[All Fields] OR "adolescent s"[All Fields]) OR ("adolescent"[MeSH Terms] OR "adolescent"[All Fields] OR "youth"[All Fields] OR "youths"[All Fields] OR "youth s"[All Fields]) OR ("adolescent"[MeSH Terms] OR "adolescent"[All Fields] OR "teenage"[All Fields] OR "teenager"[All Fields] OR "teenagers"[All Fields] OR "teenaged"[All Fields] OR "teenager s"[All Fields] OR "teenages"[All Fields]) OR ("child, preschool"[MeSH Terms] OR ("child"[All Fields] AND "preschool"[All Fields]) OR "preschool child"[All Fields] OR ("preschool"[All Fields] AND "child"[All Fields])))) AND ("pulmonary disease, chronic obstructive"[MeSH Terms] OR ("pulmonary"[All Fields] AND "disease"[All Fields] AND "chronic"[All Fields] AND "obstructive"[All Fields]) OR "chronic obstructive pulmonary disease"[All Fields] OR ("chronic"[All Fields] AND "obstructive"[All Fields] AND "pulmonary"[All Fields] AND "disease"[All Fields]) OR ("lung diseases, obstructive"[MeSH Terms] OR ("lung"[All Fields] AND "diseases"[All Fields] AND "obstructive"[All Fields]) OR "obstructive lung diseases"[All Fields] OR ("obstructive"[All Fields] AND "airway"[All Fields] AND "disease"[All Fields]) OR "obstructive airway disease"[All Fields] OR ("airflow"[All Fields] OR "airflows"[All Fields]) AND ("obstruct"[All Fields] OR "obstructed"[All Fields] OR "obstructing"[All Fields] OR "obstruction"[All Fields] OR "obstructions"[All Fields] OR "obstructive"[All Fields] OR "obstructs"[All Fields])) OR ("bronchitis, chronic"[MeSH Terms] OR ("bronchitis"[All Fields] AND "chronic"[All Fields]) OR "chronic bronchitis"[All Fields] OR ("chronic"[All Fields] AND "bronchitis"[All Fields])) OR ("emphysema"[MeSH Terms] OR "emphysema"[All Fields] OR "emphysemas"[All Fields] OR "pulmonary emphysema"[MeSH Terms] OR ("pulmonary"[All Fields] AND "emphysema"[All Fields]) OR "pulmonary emphysema"[All Fields]))	64
1+2+7		((("yoga"[MeSH Terms] OR "yoga"[All Fields] OR "meditate"[All Fields] OR "meditated"[All Fields] OR "meditating"[All Fields] OR "meditation"[MeSH Terms] OR "meditation"[All Fields] OR "meditations"[All Fields] OR "meditation s"[All Fields] OR "meditational"[All Fields] OR "meditative"[All Fields] OR "meditator"[All Fields] OR "meditators"[All Fields] OR "asana"[All Fields] OR "asanas"[All Fields] OR "pranayama"[All Fields] OR "pranayamas"[All Fields] OR "mindful"[All Fields] OR "mindfulness"[MeSH Terms] OR "mindfulness"[All Fields]) AND ("paediatrics"[All Fields] OR "pediatrics"[MeSH Terms] OR "pediatrics"[All Fields] OR "paediatric"[All Fields] OR "pediatric"[All Fields] OR ("child"[MeSH Terms] OR "child"[All Fields] OR "children"[All Fields] OR "child s"[All Fields] OR "children s"[All Fields] OR "childrens"[All Fields] OR "childs"[All Fields]) OR ("adolescences"[All Fields] OR "adolescence"[All Fields] OR "adolescent"[MeSH Terms] OR "adolescent"[All Fields] OR "adolescence"[All Fields] OR "adolescents"[All Fields] OR "adolescent s"[All Fields]) OR ("adolescent"[MeSH Terms] OR "adolescent"[All Fields] OR "youth"[All Fields] OR "youths"[All Fields] OR "youth s"[All Fields]) OR ("adolescent"[MeSH Terms] OR "adolescent"[All Fields] OR "teenage"[All Fields] OR "teenager"[All Fields] OR "teenagers"[All Fields] OR "teenaged"[All Fields] OR "teenager s"[All Fields] OR "teenages"[All Fields]) OR ("child, preschool"[MeSH Terms] OR ("child"[All Fields] AND "preschool"[All Fields]) OR "preschool child"[All Fields] OR ("preschool"[All Fields] AND "child"[All Fields])))) AND ("rhinitis, allergic"[MeSH Terms] OR ("rhinitis"[All Fields] AND "allergic"[All Fields]) OR "allergic rhinitis"[All Fields] OR ("allergic"[All Fields] AND "rhinitis"[All Fields]) OR	8

Contd...

Supplementary Table 1: Contd...

Serial number	Data base	Search strategy	Result
1+2+8		("rhinitis, allergic, seasonal"[MeSH Terms] OR ("rhinitis"[All Fields] AND "allergic"[All Fields] AND "seasonal"[All Fields]) OR "seasonal allergic rhinitis"[All Fields] OR ("hay"[All Fields] AND "fever"[All Fields]) OR "hay fever"[All Fields]) OR (("perennial"[All Fields] OR "perennialism"[All Fields] OR "perenniality"[All Fields] OR "perennially"[All Fields] OR "perennials"[All Fields] OR "perennis"[All Fields]) AND ("rhinitis"[MeSH Terms] OR "rhinitis"[All Fields] OR "rhinitides"[All Fields])) OR (("chronic"[All Fields] OR "chronical"[All Fields] OR "chronically"[All Fields] OR "chronicities"[All Fields] OR "chronicity"[All Fields] OR "chronicization"[All Fields] OR "chronics"[All Fields]) AND ("rhinitis"[MeSH Terms] OR "rhinitis"[All Fields] OR "rhinitides"[All Fields]))	19
1	EMBASE	"yoga"/exp OR yoga OR meditation OR asanas OR pranayama	25,387
2		"pediatrics"/exp OR pediatrics OR child OR "preschool child" OR adolescent	4,713,732
3		"asthma"/exp OR asthma OR bronchoconstriction	395,191
4		"lung tuberculosis"/exp OR "lung tuberculosis" OR bacteremia	161,854
5		"cystic fibrosis"/exp OR "cystic fibrosis"	111,074
6		"chronic obstructive lung disease"/exp OR "chronic obstructive lung disease" OR bronchitis OR "lung emphysema"	250,997
7		"allergic rhinitis"/exp OR "allergic rhinitis"	64,172
8		"pneumonia"/exp OR pneumonia OR "pleura effusion"	540,275
1+2			2801
1+2+3			107
1+2+4			2

Contd...

Supplementary Table 1: Contd...

Serial number	Data base	Search strategy	Result
1+2+5			27
1+2+6			15
1+2+7			8
1+2+8			15
1	SCOPUS	(ALL (yoga) OR ALL (asana) OR ALL (pranayama) OR ALL (meditation))	158,191 results
2		(ALL (pediatrics) OR ALL (child) OR ALL (child AND pre AND school) OR ALL (adolescent))	10,026,506
3		(ALL (asthma) OR ALL (bronchial AND asthma) OR ALL (bronchoconstriction))	774,970 results
4		(ALL (tuberculosis) OR ALL (lung AND tuberculosis) OR ALL (bacteremia))	932,879
5		(ALL (cystic AND fibrosis) OR ALL (mucoviscidosis))	267,564
6		ALL (allergic AND rhinitis)	115,383
7		(ALL (chronic AND obstructive AND pulmonary AND disease) OR ALL (chronic AND airway AND disease) OR ALL (bronchitis) OR TITLE-ABS-KEY (pulmonary AND emphysema))	590,436
8		(ALL (pneumonia) OR ALL (pleural AND effusion))	1,086,702
9 (1+2+3)		((ALL (yoga) OR ALL (asana) OR ALL (pranayama) OR ALL (meditation))) AND ((ALL (pediatrics) OR ALL (child) OR ALL (child AND pre AND school) OR ALL (adolescent))) AND ((ALL (asthma) OR ALL (bronchial AND asthma) OR ALL (bronchoconstriction)))	2557
1+2+4		((ALL (yoga) OR ALL (asana) OR ALL (pranayama) OR ALL (meditation))) AND ((ALL (pediatrics) OR ALL (child) OR ALL (child AND pre AND school) OR ALL (adolescent))) AND ((ALL (tuberculosis) OR ALL (lung AND tuberculosis) OR ALL (bacteremia)))	552
1+2+8		((ALL (yoga) OR ALL (asana) OR ALL (pranayama) OR ALL (meditation))) AND ((ALL (pediatrics) OR ALL (child) OR ALL (child AND pre AND school) OR ALL (adolescent))) AND ((ALL (pneumonia) OR ALL (pleural AND effusion)))	690
1+2+5		((ALL (yoga) OR ALL (asana) OR ALL (pranayama) OR ALL (meditation))) AND ((ALL (pediatrics) OR ALL (child) OR ALL (child AND pre AND school) OR ALL (adolescent))) AND ((ALL (cystic AND fibrosis) OR ALL (mucoviscidosis)))	473
1+2+6		((ALL (yoga) OR ALL (asana) OR ALL (pranayama) OR ALL (meditation))) AND ((ALL (pediatrics) OR ALL (child) OR ALL (child AND pre AND school) OR ALL (adolescent))) AND (ALL (allergic AND rhinitis))	288